

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0537

Coronado 345 kV
March 04, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0537. The Interconnection Customer proposes a 258.5 MW solar photovoltaic facility interconnecting at the Coronado 345 kV substation in the New Jersey area. The request was received on November 12, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 265 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$114,146,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CABINET GORGE - WALNUT-ENGLE TAP 138 KV	ORANGE - MOJAVE SIPHON 345 KV	158.6	254	6.7
MOJAVE SIPHON - ROOSEVELT DAM 138 KV L	ORANGE - MOJAVE SIPHON 345 KV	154.6	259	8.6
ROOSEVELT DAM - BULLOCK 138 KV	MOJAVE SIPHON - ORANGE 345 KV Line	154.5	261	8.6
BULLOCK - ARROWHEAD 138 KV line	ORANGE - MOJAVE SIPHON 345 KV Line	151.3	259	8.6
ARROWHEAD - HARRIS BRANCH 138 KV	ORANGE - MOJAVE SIPHON 345 KV	141.0	246	8.6
MOJAVE SIPHON - CORONADO 345 KV	BROWNFIELD CITY - CORONADO 500 KV	138.3	1,204	54.3
HARRIS BRANCH - ATCHELITZ 138 KV	MOJAVE SIPHON SUBSTATION - ORANGE SUBSTATION 345 KV	137.8	242	8.6
WALNUT-ENGLE TAP - ATCHELITZ 138 KV	MOJAVE SIPHON - ORANGE 345 KV	133.3	226	8.6
ORANGE - MOJAVE SIPHON 345 KV	BROWNFIELD CITY - CORONADO 500 KV	130.6	1,009	50.3
GRANDVIEW - CORONADO 138 KV	MOJAVE SIPHON - CORONADO 345 KV L	129.3	243	7.6
ROSITA CREEK - GRANDVIEW 138 KV	MOJAV SIPHON - CORONADO 345 KV	125.3	237	7.6
GATEWAY - ROSITA CREEK 138 KV L	COONADO - MOJAVE SIPHON SUBSTATION 345 KV	122.1	227	7.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GATEWAY - BLANCHARDVILLE SUBSTATION 138 KV	MOJAVE SIPHON - CORONADO 345 KV Line	120.2	220	7.6
GLENMORE - BLANCHARDVILLE 138 KV	CORONADO - MOJAVE SIPHON 345 KV	118.4	213	7.6
HAMILTON DOME - OAK KNOLL 138 KV	MJAVE SIPHON - CORONADO SUBSTATION 345 KV	100.9	299	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WALNUT-ENGLE TAP - CABINET GORGE 138 KV L	Rebuild line with higher-capacity conductor	\$3,230,000	100% (\$3,230,000)
ROOSEVELT DAM - MOJAVE SIPHON 138 KV	Rebuild line with higher-capacity conductor	\$683,000	100% (\$683,000)
ROOSEVELT DAM - BULLOCK 138 KV	Rebuild line with higher-capacity conductor	\$6,669,000	100% (\$6,669,000)
BULLOCK - ARROWHEAD 138 KV line	Rebuild line with higher-capacity conductor	\$3,072,000	100% (\$3,072,000)
ARROWHEAD - HARRIS BRANCH 138 KV	Rebuild line with higher-capacity conductor	\$4,298,000	100% (\$4,298,000)
CORONADO - MOJAVE SIPHON 345 KV	Rebuild line with higher-capacity conductor	\$87,515,000	50% (\$43,610,000)
HARRIS BRANCH - ATCHELITZ 138 KV	Rebuild line with higher-capacity conductor	\$4,184,000	100% (\$4,184,000)
ATCHELITZ - WALNUT-ENGLE TAP 138 KV	Rebuild line with higher-capacity conductor	\$7,536,000	100% (\$7,536,000)
ORANGE - MOJAVE SIPHON 345 KV	Rebuild line with higher-capacity conductor	\$17,876,000	53% (\$9,454,000)
CORONADO - GRANDVIEW 138 KV L	Rebuild line with higher-capacity conductor	\$1,979,000	100% (\$1,979,000)
ROSITA CREEK - GRANDVIEW 138 KV	Rebuild line with higher-capacity conductor	\$8,187,000	100% (\$8,187,000)
GATEWAY - ROSITA CRK 138 KV	Rebuild line with higher-capacity conductor	\$8,647,000	100% (\$8,647,000)
GATEWAY - BLANCHARDVILLE SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$6,379,000	100% (\$6,379,000)
GLENMORE - BLANCHARDVILLE 138 KV	Rebuild line with higher-capacity conductor	\$5,371,000	100% (\$5,371,000)
OAK KNOLL - HAMILTON DOME 138 KV	Reconductor line	\$847,000	100% (\$847,000)
	Total allocated to project		\$114,146,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.